# SDC

# Hackathon · Phase 1

Preliminary Project Proposal

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**Lumina Nusantara** · Domain 1 — The Aiversecity Future-Ready Workforce Domain

SDG Track: SDG 4 — Quality Education (sub-target 4.4, with 8.5)

Team Leader: DU YULIN · Xiamen University Malaysia (XMUM)

Live prototype: https://sdc2026-lumina-nusantara-65cnugvuxdle5dgp9ky3fs.streamlit.app

Code repository: https://github.com/douyinlin/sdc2026-lumina-nusantara

**Submission deadline: 25 October 2026**

Phase 1 — Preliminary Project Proposal. Answers below follow the official template order and the stated word limits (2.1 ≤ 250 words, 2.2 ≤ 300 words, 3.2 ≤ 200 words).

## 1. TEAM AND PROJECT OVERVIEW

| | |
|---|---|
| **Team Name** | Lumina Nusantara |
| **Selected Domain** | Domain 1 — The Aiversecity Future-Ready Workforce Domain |
| **Selected SDG Track (1–17)** | SDG 4 — Quality Education (sub-target 4.4, with 8.5) |
| **Project / Solution Name** | SkillRadar MY |
| **Name (Leader)** | DU YULIN |
| **Email & Phone (Leader)** | CYS2609009@xmu.edu.my / +60 11-1798 1756 |
| **University Affiliation(s)** | Xiamen University Malaysia (XMUM) |
| **Number of Team Members** | 5 |

## 2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

### 2.1 The Problem Context

Malaysia's labour market is being reshaped faster than its training system can respond. 56% of Malaysian professionals already use AI tools at work, yet only 26% have received formal training in them (Hays Asia Salary Guide 2025), and the Ministry of Science, Technology and Innovation expects more than 600,000 workers to need reskilling within three to five years.

The people most exposed are rarely the loudest in this debate: a 34-year-old customer-service supervisor in Klang whose tasks are being automated one by one; a TVET graduate applying for entry roles that no longer exist; an L&D manager in a 200-person SME who must set next year's training budget without evidence of which roles will still be viable.

Three failures repeat. Workers cannot see which skills are rising, so they buy credentials that are already commoditised. Employers cannot translate AI trends into a training plan, so budgets default to generic soft-skills courses. And career guidance is a snapshot: a one-off skills assessment in a market that keeps moving, so the advice expires before the learner finishes the course.

The cost is economic and social — wasted tuition, slower wage growth, and widening gaps for older workers, women returning from care work, and rural learners who cannot reach premium bootcamps. Without a system that updates as AI advances and shows how each role connects to the SDGs, reskilling stays an expensive guess rather than a pathway out.

### 2.2 The Proposed AI Solution

**Intake.** A user answers a short questionnaire on role, years of experience, tools and interests, or uploads a CV that we parse into a normalised skill profile. Every skill maps to a shared taxonomy (O*NET / ESCO identifiers) through an alias dictionary, so "Zendesk" and "crm" both resolve to CRM administration.

**Trend layer.** A scheduled job ingests public signals — WEF Future of Jobs indicators, O*NET skill updates, sampled public job postings and the HRD Corp course catalogue — into a versioned trend snapshot. Each record stores its source URL, retrieval date and metric, and the interface exposes that provenance on every forecast, so anyone can see when the data changed and what moved.

**Matching.** For each candidate future role we compute an explainable score from four visible components: skill coverage, transferability of adjacent skills, demand trend and gap cost. The user sees the arithmetic and the exact missing skills, not a black box.

**Pathway and scenarios.** Missing skills are ordered cheapest-first into a week-by-week plan built from free or low-cost resources, each with a link and estimated hours. The ranking runs twice — moderate and fast AI disruption — so the user sees how robust a choice is. Every role and skill carries SDG tags (primary 4.4 and 8.5, with 8.2 and 9.1 where relevant).

**From the Phase 1 workshops** we apply GenAI APIs for CV parsing and plain-language explanations, a relational schema for profiles, trends and courses, and cloud deployment on Replit and Streamlit Community Cloud, so the prototype is genuinely online rather than a slide.

## 3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

### 3.1 Conceptual Tech Stack

| Layer | Technology | Why |
|---|---|---|
| UI | Streamlit (Python) | Fast iteration, runs unchanged on Replit and Streamlit Cloud |
| Service layer | Python modules (src/utils.py) | Testable matching / pathway / SDG logic |
| Data | SQLite locally, Postgres on hosted deployment | Versioned trend snapshots, profiles, courses |
| AI | No language model at inference time; deterministic rule engine | GPU embeddings for CV parsing; LLM only phrases explanations |
| Ingestion | Python scheduled job, JSON snapshots | Each snapshot carries source_url + retrieved_at |
| Hosting | Replit; Streamlit Community Cloud | Public live demo URL for judges |
| Tests | python -m unittest discover tests | Regression on score and pathway logic |

**System block diagram** — see the appended diagram (*docs/architecture.png* in the repository).

> User → Streamlit UI → Matching & pathway engine → SQLite/Postgres; Ingestion job → trend_snapshots(source_url, retrieved_at); Streamlit UI reads the trend snapshot; all parsing and scoring run locally.

**Sponsor-provided GenAI API or Cloud Platform:** Replit only. The prototype was developed in a Replit workspace. **No RunPod GPU endpoint is called by the application** — CV parsing runs on a local rule-based parser, and every score comes from a deterministic rule set. The public live demo is hosted on Streamlit Community Cloud from the same repository.

### 3.2 Design-Thinking & Deployment Strategy

**Empathise and define.** Five semi-structured interviews — one HR executive, one TVET lecturer, two working adults considering a switch, one final-year student — plus a review of Malaysian reskilling reports. The recurring sentence was "I don't know what to learn next, and I don't trust course adverts". We defined the job to be done as: in under ten minutes, understand which two or three roles are realistic for me and what the first step is.

**Ideate and prototype.** Paper flows, then a clickable Streamlit prototype tested with the same five people; we cut features that needed data we could not source honestly.

**Test and de-risk.** The score is transparent by construction, every claim links to a dated source, and parsing runs locally with no external AI service so a live demo cannot fail on stage.

**Deploy.** Replit is the shared workspace and Streamlit Community Cloud hosts the live demo; a weekly snapshot job keeps trends current. Phase 2 adds the HR view and automatic re-planning.

## 4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

> Baselines are taken from published reports or our own five-person interviews. The interview count is stated so judges can weigh the evidence.

| Metric Category | Your Response |
|---|---|
| Target SDG Sub-Target(s) | **4.4** — substantially increase the number of youth and adults who have relevant skills, including technical and vocational skills, for employment and entrepreneurship. **8.5** — achieve full and productive employment and decent work for all. Secondary: **4.3** equal access to affordable technical and vocational training. |
| Current Problem Baseline | 56% of Malaysian professionals use AI tools at work but only **26%** have received formal training (Hays Asia Salary Guide 2025); **600,000+** workers need reskilling within 3–5 years (MOSTI). A learner currently spends 3–8 weeks (median 4) on self-directed search across job boards and course catalogues before committing to a route (our five interviews). |
| Core Measurable KPI | (1) **Minutes from profile to a ranked, sourced reskilling plan** — target ≤5 min, measured in-app. (2) **Top-3 relevance rate** — share of pilot users rating at least one of the three recommended roles as realistic (5-point scale ≥4). (3) **Source coverage** — share of displayed recommendations carrying a source URL and retrieval date; must be 100%. |
| Phase 2 Impact Target | **70%** top-3 relevance across ≥30 pilot users; ≥**40%** reduction in time-to-plan versus the 4-week baseline; **100%** of role and skill cards traceable to a dated public source; ≥**60%** of pilot users complete step 1 of their pathway within 30 days. |

**Sources:** Hays Asia Salary Guide 2025; Ministry of Science, Technology and Innovation (MOSTI) reskilling estimate as cited in the SDC Domain 1 brief; WEF Future of Jobs Report; O*NET database (U.S. Department of Labor); HRD Corp course directory.

## 5. ELIGIBILITY & READINESS CHECKLIST

■ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

■ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs (SDG 4.4 and 8.5).

■ The proposed solution targets a functional, deployed application rather than a theoretical business case study — live at https://sdc2026-lumina-nusantara-65cnugvuxdle5dgp9ky3fs.streamlit.app.

■ All fields in this template have been completed within the word limits stated.

## 6. IP & ORIGINALITY DECLARATION

### 6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon? **No.** All code, design and research was created during the SDC 2026 hackathon period.

### 6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create? **Yes** — declared below.

| Name | Type (code/dataset/model) | License |
|---|---|---|
| Streamlit | code | Apache 2.0 |
| pandas | code | BSD 3-Clause |
| O*NET database (U.S. Dept. of Labor) | dataset | CC BY 4.0 (attributed in-app) |
| WEF Future of Jobs indicators | dataset | Public report, cited |
| HRD Corp course directory | dataset | Public directory, cited |

| Name | Type (code/dataset/model) | License |
|---|---|---|
| O*NET occupation skill lists | dataset | CC BY 4.0 |
| Streamlit | code | Apache 2.0 |

### 6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon? **No.**

## 7. DECLARATION

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

Signature: 

Name: DU YULIN

Date: _____ / _____ / 2026
25 / 10 / 2026